FINITE INDEX OF ITERATED GALOIS GROUPS AT STRICTLY PREPERIODIC POINTS FOR UNICRITICAL PRIME-POWER POLYNOMIALS

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Allowed standard prerequisites (editorial).

- Original arithmetic geometry and number theory prerequisites. Original complete bibliography and source-local definitions retained.

1. Introduction and Statement of Results

Let K be a field. Let $f \in K(x)$ with $d = \deg f \geq 2$ and let $\beta \in K$. For $n \in \mathbb{N}$, let $K_n(\beta) = K(f^{-n}(\beta))$ denote the field obtained by adjoining the nth preimages of β under f to $K(\beta)$. (We declare that $K_0(\beta) = K$.) Set $K_\infty(\beta) = \bigcup_{n=1}^{\infty} K_n(\beta)$. For $n \in \mathbb{N} \cup \{\infty\}$, define $G_n(\beta) = \mathrm{Gal}(K_n(\beta)/K(\beta))$, and let $G_\infty(\beta) = \varprojlim G_n(\beta)$ (note that there are natural projection maps $G_i(\beta) \longrightarrow G_j(\beta)$ for $i > j$).

When β is not in the forward orbit of any critical point. the group $G_\infty(\beta)$ embeds into $\mathrm{Aut}(T_\infty^d)$, the automorphism group of an infinite d-ary rooted tree T_∞^d. Recently there has been much work on the problem of determining when the index $[\mathrm{Aut}(T_\infty^d) : G_\infty(\beta)]$ is finite. The group $G_\infty(\beta)$ is the image of an arboreal Galois representation, so this finite index problem is an analog in arithmetic dynamics of the finite index problem for the ℓ-adic Galois representations associated to elliptic curves, resolved by Serre's celebrated Open Image Theorem [Ser72]. By work of Odoni [Odo85], one expects that many rational functions have a surjective arboreal representation, i.e., that $[\mathrm{Aut}(T_\infty^d) : G_\infty(\beta)] = 1$ (see also [BJ19, Loo19, Kad20, DK22, Jon13]).

For special rational functions (such as $f(x) = x^d + c$ for $d \geq 3$), $G_\infty(\beta)$ will be much smaller then $\mathrm{Aut}(T_\infty^d)$, but one can generally still expect $G_\infty(\beta)$ to have finite index inside a group that is naturally associated to f in many cases. Indeed, if we let $\mathbf{G}_n$ denote the Galois group $\mathrm{Gal}(K(f^{-n}(t))/K(t))$ for t transcendental over K and let $\mathbf{G}_\infty$ be the inverse limit of the $\mathbf{G}_n$, we have a candidate for such a group. The following treats a special case of a more general question from [BDG+21b]. We restrict here to the case where f is a polynomial, because the conjecture is slightly more complicated for general rational functions.

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Question 1.1. *Let f be a polynomial that is not conjugate to a monomial or Chebychev polynomial and let β be a point that is not periodic or post-critical for f. Is it true that we must have $[\mathbf{G}_\infty : G_\infty(\beta)] < \infty$?*

It is easy to see that $G_\infty(\beta)$ cannot have finite index in $\mathbf{G}_\infty$ when β is periodic since the number of irreducible factors of $f^n(x) - \beta$ goes to infinity as n goes to infinity. Likewise, when β is post-critical, one cannot form the tree of iterated inverse images of β in the usual way. Thus, this question asks if we must have $[\mathbf{G}_\infty : G_\infty(\beta)] < \infty$ except in the case of the obvious exceptions. Recently, Benedetto and Jones [BJ25] have shown that there are PCF quadratic rational functions and β that are neither periodic nor post-critical such that $\mathbf{G}_\infty : [G_\infty(\beta)] = \infty$, so Question 1.1 does not always have a positive answer.

In this paper we study the family of polynomials $f(x) = x^d + c$ for $c \in K$, which up to change of variables represents all polynomials with precisely one (finite) critical point. If K contains a primitive d-th root of unity ξ_d, then it is easy to see that $\mathbf{G}_n$ is isomorphic to $[C_d]^n$, so that $\mathbf{G}_\infty$ is isomorphic to the infinite iterated wreath product $[C_d]^\infty$. For more general K, we see that $\mathbf{G}_n$ will contain $[C_d]^n$ as a subgroup of index $[K(\xi_d) : K]$ where ξ_d is a primitive d-th root of unity. Iterated Galois groups of polynomials of the form $x^d + c$ hav been studied extensively, see [Jon07, Jon08, HJM15], for example.

Before stating our main results, we set some notation. Throughout this paper, unless otherwise indicated, K will refer to a number field. We say $\beta \in \overline{K}$ is periodic for f if $f^n(\beta) = \beta$ for some $n \geq 1$, and we say β is preperiodic for f if $f^m(\beta)$ is periodic for some $m \geq 0$. We say that β is **strictly preperiodic** for f it is preperiodic for f but not periodic for f. A rational function f is said to be post-critically finite, or **PCF**, if all of its critical points are preperiodic under f.

Our main result is a finite index statement for iterated Galois groups of strictly preperiodic points for polynomials of the form $f(x) = x^q + c$ where q is a prime power and 0 is not preperiodic under f (so that f is not PCF).

Theorem 1.2. *Let K be a number field. Let $f(x) = x^q + c \in K[x]$, where c is an algebraic integer, $q = p^r$ is a power of a prime number p, and 0 is not preperiodic for f. Let $\beta \in \overline{K}$ be strictly preperiodic for f. Then*

$$[\mathbf{G}_\infty : G_\infty(\beta)] < \infty.$$

The condition on the degree of f ensures that the pair (f, β) is eventually stable (see Definition 2.3), a necessary condition that is difficult to verify in general but is known to hold in this case via work of [JL17] (see Theorem 5.2). The proof can be made to work for more general non-PCF $x^d + c$ under the assumption of eventual stability, but the proof is a bit more complicated. The condition that β be strictly preperiodic is what allows us to apply Proposition 3.1 in place of the diophantine conjectures

such as Vojta's conjeture and the *abc*-conjecture that are often used to prove conditional finite index results.

We can also prove a disjointness result for fields generated by inverse images of distinct preperiodic points.

Theorem 1.3. *Let K be a number field. Let $f(x) = x^q + c \in K[x]$ where c is an algebraic integer $q = p^r$ is a power of a prime number p, and 0 is not preperiodic for f. Let $\beta_1, \dots, \beta_t \in \overline{K}$ be strictly preperiodic under f and suppose that there are no distinct j, k with the property that $f^m(\beta_j) = \beta_k$ for some $m > 0$. For each j, let M_j denote $K_\infty(\beta_j)$. Then for each $j = 1, \dots, t$, we have*

$$\left[M_j \cap \left(\prod_{k \neq j} M_k \right) : K \right] < \infty.$$

Both Theorems 1.2 and 1.3 hold under slightly weaker conditions; see Theorems 5.3 and 5.4.

Theorem 1.3 also has a natural interpretation as a finite index result across pre-image trees of several points (see Section 6). This allows us to state a theorem about iterated Galois groups of periodic points as well. The statement here is slightly more complicated, though it says essentially the same thing as Theorem 1.2, namely that the iterated Galois group has finite index in a certain "largest possible group"; see Remark 6.2.

The technique of the proof is very similar to that of [BT19, BDG+21a]. The two main differences are that we already have eventual stability by work of [JL17] and that we may use Proposition 3.1 in place of more complex diophantine arguments in order to produce primitive divisors. Proposition 3.1 produces a slightly weaker condition on our primitive prime divisors that requires some small changes in the Galois theoretic arguments of Section 4.

An outline of the paper is as follows. In Section 2, we introduce some background on wreath products and irreducibility of iterates of polynomials. In Section 3, we prove our main diophantine result, Proposition 3.1, which guarantees the existence of primes $\mathfrak{p}$ such that $v_\mathfrak{p}(f^n(0) - \beta)$ is positive and prime to p for all sufficiently large n; we note that it is crucial here that β be strictly preperiodic. Following that, in Section 4, we prove introduce Condition R (adapted from [BT19]), and prove results showing that $\mathrm{Gal}(K_n(\beta)/K_{n-1}(\beta))$ is large as possible when this condition are satisfied at β for f and n. Then, in Section 5, we combine Proposition 3.1 with the results of Section 4 to prove Theorem 5.3 and Theorem 5.4, are slight generalizations of Theorem 1.2 and Theorem 1.3. Finally, in Section 6, we introduce the multitree associated to our points, which allows us to phrase a finite index result for several points at once.

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2. Preliminaries

We will gather a few basic results about wreath products and irreducibility of polynomials.

2.1. **Wreath products.** Let G be a permutation group acting on a set X, and let H be any group. Let H^X be the group of functions from X to H with multiplication defined pointwise, or equivalently the direct product of $|X|$ copies of H. The *wreath product* of G by H is the semidirect product $H^X \rtimes G$, where G acts on H^X by permuting coordinates: for $f \in H^X$ and $g \in G$ we have

$$f^g(x) = f(g^{-1}x)$$

for each $x \in X$. We will use the notation $G[H]$ for the wreath product, suppressing the set X in the notation. (Another common convention is $H \wr G$ or $H \wr_X G$ if we wish to call attention to X.)

Fix an integer $d \geq 2$. For $n \geq 1$, let T_n^d be the complete rooted d-ary tree of level n. It is easy to see that $\mathrm{Aut}(T_1^d) \cong S_d$, and standard to show that $\mathrm{Aut}(T_n^d)$ satisfies the recursive formula

$$\mathrm{Aut}(T_n^d) \cong \mathrm{Aut}(T_{n-1}^d)[S_d].$$

Therefore we may think of $\mathrm{Aut}(T_n^d)$ as the "nth iterated wreath product" of S_d, which we will denote $[S_d]^n$. In general, for $f \in K[x]$ of degree d and $\beta \in K$, the Galois group $G_n(\beta) = \mathrm{Gal}(K_n(\beta)/K)$ embeds into $[S_d]^n$ via the faithful action of $G_n(\beta)$ on the nth level of the tree of preimages of β (see for example [Odo85] or [BT19, Section 2]).

Assume now that $f(x) := x^d + c \in K[x]$, where K is a field of characteristic 0 that contains the dth roots of unity. For $\beta \in K$ such that $\beta - c$ is not a dth power in K, we have $K_1(\beta) = K((\beta - c)^{1/d})$ and $G_1(\beta) \cong C_d$. For any $n \geq 2$, the extension $K_n(\beta)$ is a Kummer extension attained by adjoining to $K_{n-1}(\beta)$ the dth roots of $z - c$ where z ranges over the roots of $f^{n-1}(x) = \beta$. Thus we have

$$\mathrm{Gal}(K_n(\beta)/K_{n-1}(\beta)) \subseteq \prod_{f^{n-1}(z)=\beta} \mathrm{Gal}(K_{n-1}(\beta)((z-c)^{1/d})/K_{n-1}(\beta)) \subseteq C_q^{q^{n-1}}.$$

This is clear if $f^{n-1}(x) - \beta$ has distinct roots in $\overline{K}$. If $f^{n-1}(x) - \beta$ has repeated roots, then $\mathrm{Gal}(K_n(\beta)/K_{n-1}(\beta))$ sits inside a direct product of a smaller number of copies of C_d, so the stated containments still hold.

Considering the Galois tower

$$K_n(\beta) \supseteq K_{n-1}(\beta) \supseteq K$$

we see that

$$G_n(\beta) \subseteq \mathrm{Gal}(K_n(\beta)/K_{n-1}(\beta)) \rtimes G_{n-1}(\beta) \cong G_{n-1}(\beta)[C_d],$$

where the implied permutation action of $G_{n-1}(\beta)$ is on the set of roots of $f^{n-1}(x) - \beta$. By induction, $G_n(\beta)$ embeds into $[C_d]^n$, the nth iterated wreath product of C_d. Observe that $[C_d]^n$ sits as a subgroup of $\mathrm{Aut}(T_n^d) \cong [S_d]^n$ via

the obvious action on the tree. Taking inverse limits, $G_\infty(\beta)$ embeds into $[C_d]^\infty$, which sits as a subgroup of $\mathrm{Aut}(T_\infty)$.

We summarize our basic strategy for proving that $G_\infty(\beta)$ has finite or infinite index in $[C_d]^\infty$ as Proposition 2.1.

Proposition 2.1. *Let $f = x^d + c \in K[x]$. Then $[\mathbf{G}_\infty : G_\infty(\beta)] < \infty$ if and only if $\mathrm{Gal}(K_n(\beta)/K_{n-1}(\beta)) \cong C_q^{q^{n-1}}$ for all sufficiently large n.*

Proof. Since $[K(\xi_d) : K]$ is finite, it suffices to prove this when K contains ξ_d. Thus, we may assume that $\mathbf{G}_\infty$ is $[C_d]^\infty$. Consider the projection map $\pi_n : [C_d]^\infty \to [C_d]^n$. The restriction of π_n maps $G_\infty(\beta)$ to $G_n(\beta)$. By basic group theory, we have

$$[[C_d]^\infty : G_\infty(\beta)] \geq [[C_d]^n : G_n(\beta)].$$

Therefore if $\mathrm{Gal}(K_n(\beta)/K_{n-1}(\beta))$ is a proper subgroup of $C_q^{q^{n-1}}$ for infinitely many n, then $[[C_d]^n : G_n(\beta)]$ is unbounded as $n \to \infty$, and $[[C_d]^\infty : G_\infty(\beta)] = \infty$.

Conversely, by appealing to the profinite structure of $[C_d]^\infty$ we see that distinct cosets of $G_\infty(\beta)$ in $[C_d]^\infty$ must project to distinct cosets in $[C_d]^n$ under π_n for some n. If there exists N such that $\mathrm{Gal}(K_n(\beta)/K_{n-1}(\beta)) \cong C_q^{q^{n-1}}$ for all $n > N$, then by induction,

$$[[C_d]^n : G_n(\beta)] \leq [[C_d]^N : G_N(\beta)]$$

for all n. Thus $[[C_d]^\infty : G_\infty(\beta)] \leq [[C_d]^N : G_N(\beta)]$ as well. □

2.2. **Capelli's lemma and eventual stability.** We will use Capelli's lemma throughout this paper. The lemma is standard (see [Odo88] or [BT19, Lemma 4.1], for example. We state it here without proof.

Lemma 2.2 (Capelli's Lemma)**.** *Let K be any field and let $f, g \in K[x]$. Suppose $\alpha \in \overline{K}$ is any root of f. Then $f(g(x))$ is irreducible over K if and only if both $f(x)$ is irreducible over K and $g(x) - \alpha$ is irreducible over $K(\alpha)$.*

Definition 2.3. Let K be a number field, let f be a rational function, and let $\beta \in \mathbb{P}^1(K)$. We say that the pair (f, β) is **eventually stable over** K if there is a constant C such for any n, the the number of $\mathrm{Gal}(\overline{K}/K)$-orbits of points in $f^{-n}(\beta)$ is less than C. (Note that C depends on K, f, and β in general.)

When f is a polynomial and β is not the point at infinity this is equivalent to saying that the number of irreducible factors of $f^n(x) - \beta$ over $K[x]$ is bounded independently of n.

The following is a simple application of Capelli's lemma (see [BT19, Proposition 4.1]).

Proposition 2.4. *Let K be a number field, let $f \in K[x]$, and let $\beta \in K$. If the pair (f, β) is eventually stable over K, there exists some $N \geq 0$ such that for every element of $\alpha \in f^{-N}(\beta)$, the polynomial $f^n(x) - \alpha$ is irreducible over $K(\alpha)$ for all $n \geq 0$.*

3. Primitive prime divisors

Proposition 3.1 is the main diophantine tool used in this paper. The proof is similar to that of [BIJ+17, Proposition 12]. It provides us with primitive prime divisors of $f^n(0)$, that is prime divisors $\mathfrak{p}$ of $f^n(0)$ that are not divisors of $f^m(0)$ for any $m < n$. In general, a prime $\mathfrak{p}$ in a number field K is said to be primitive prime divisor of a_n for a_n an element of sequence (a_i) of elements of K if $v_\mathfrak{p}(a_n) > 0$ and $v_\mathfrak{p}(a_m) \leq 0$ for all $m < n$.

We say that a polynomial

$$f(x) = a_d x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0,$$

has **good reduction** at $\mathfrak{p}$ if $v_\mathfrak{p}(a_d) = 0$ and $v_\mathfrak{p}(a_i) \geq 0$ for $0 \leq i \leq d-1$. See [MS94] or [Sil07, Theorem 2.15] for a more careful definition that also applies to rational functions. Clearly, any f has good reduction at all but finitely many $\mathfrak{p}$. The idea behind the definition as used here is that if f has good reduction at $\mathfrak{p}$, then f commutes with the reduction mod $\mathfrak{p}$ map $r_\mathfrak{p} : \mathbb{P}^1(\overline{K}) \to \mathbb{P}^1(\overline{k}_\mathfrak{p})$. This is clear for polynomials (see [Sil07, Theorem 2.18] for a proof for rational functions). We say that f has **good separable reduction** at $\mathfrak{p}$ if the reduced map $\bar{f} : \mathbb{P}^1(\overline{k}_\mathfrak{p}) \to \mathbb{P}^1(\overline{k}_\mathfrak{p})$ is separable.

Proposition 3.1. *Let $d > 1$ be an integer, let K be a number field, let f be a polynomial of degree greater than 1, and let S be a finite set of primes of K. Let $\alpha \in \mathbb{P}^1(K)$ be a point that is not preperiodic for f. Let $\beta \in K$ be a point that is strictly preperiodic for f but not post-critical for f. Then for all sufficiently large n, there is a prime $\mathfrak{p} \notin S$ such that*

(1) $v_\mathfrak{p}(f^n(\alpha) - \beta)$ is positive and not divisible by d; and
(2) $v_\mathfrak{p}(f^m(\alpha) - \beta) = 0$ for all $m < n$.

Proof. Since β is strictly preperiodic, the set $\{f(\beta), \dots, f^k(\beta), \dots\}$ is finite and does not include β as an element. Thus, there are at most finitely many primes $\mathfrak{p}$ of K such that $f^k(\beta) \equiv \beta \pmod{\mathfrak{p}}$ for some $k > 0$, so after expanding S to a larger finite set, we may assume that it contains all such primes $\mathfrak{p}$. Likewise, also possibly after expanding S, we may assume that the ring $\mathfrak{o}_S$ of S-integers in K (that is elements of K that are integers at every prime outside of S) is a principal ideal domain. We may also assume that S contains all primes of bad reduction for f and all primes $\mathfrak{p}$ such that $v_\mathfrak{p}(\alpha) < 0$ or $v_\mathfrak{p}(\beta) < 0$.

Now, if $v_\mathfrak{p}(f^n(\alpha) - \beta)$ is positive for $\mathfrak{p} \notin S$, then we cannot have $v_\mathfrak{p}(f^m(\alpha) - \beta) > 0$ for any $m < n$ since if we did we would have $f^{n-m}(\beta) \equiv \beta \pmod{\mathfrak{p}}$, which is impossible since $\mathfrak{p} \notin S$. Similarly, we cannot have $v_\mathfrak{p}(f^m(\alpha) - \beta) < 0$ for any m since $v_\mathfrak{p}(f^m(\alpha))$ and $v_\mathfrak{p}(\beta)$ are both always non-negative since f has good reduction at $\mathfrak{p}$ and α and β are both integers at $\mathfrak{p}$. Thus, condition (2) above will be met whenever (1) is, so it suffices to show that there is an n_0 such that for all $n \geq n_0$, there is an $\mathfrak{p} \notin S$ such that $v_\mathfrak{p}(f^n(\alpha) - \beta)$ is positive and not divisible by d.

The ring of $\mathfrak{o}_S^*$ of S-units is a finitely generated group, so $\mathfrak{o}_S^*/(\mathfrak{o}_S^*)^d$ is a finite group. Let $\gamma_1(\mathfrak{o}_S^*)^d, \ldots, \gamma_N(\mathfrak{o}_S^*)^d$ be the set of cosets of $(\mathfrak{o}_S^*)^d$ in $\mathfrak{o}_S^*$. Since β is not post-critical, $f^3(x) - \beta$ has at least 8 and thus more than 4 roots. It follows from Riemann-Hurwitz that for each i, the curve C_i given by $y^d = \gamma_i(f^3(x) - \beta)$ has genus greater than 1. By Faltings' theorem, this means that each C_i has finitely many rational points. Thus, since α is not preperiodic there is an n_0 such that for all $n \geq n_0$ there is no $y \in K$ such that $y^d = \gamma_i(f^3(f^{n-3}(\alpha)) - \beta)$.

Now, let $n \geq n_0$ and suppose that for every prime $\mathfrak{p} \in S$, we have that $v_\mathfrak{p}(f^n(\alpha) - \beta)$ is either 0 or divisible by d. Then the $\mathfrak{o}_S$ ideal generated by $f^n(\alpha) - \beta$ is the d-th power of the $\mathfrak{o}_S$-ideal I. Let z be a generator for I as an $\mathfrak{o}_S$ ideal (such a z exists since $\mathfrak{o}_S$ is a principal ideal domain). Then we have $z^d = u(f^n(\alpha) - \beta)$ for a unit $u \in \mathfrak{o}_s^*$. We may write $u = \gamma_i w^d$ for one of our coset representatives γ_i and some unit $w \in \mathfrak{o}_S^*$. Let $y = z/w$. Then we have $y^d = \gamma_i(f^3(f^{n-3}(\alpha))) - \beta)$ with $y \in K$, a contradiction.

Thus, for every $n \geq n_0$, there is an $\mathfrak{p} \notin S$ such that $v_\mathfrak{p}(f^n(\alpha) - \beta)$ is positive and not divisible by d.

□

4. Ramification and Galois theory

Throughout this section, $f(x)$ will denote a polynomial of the form $f(x) = x^q + c$ where $c \in K$ for K a number field, the critical points 0 is not preperiodic, and $q = p^r$ is a power of a prime p.

In this section we define **Condition R** and **Condition U** in terms of primes dividing certain elements of K related to the forward orbits of 0. In Proposition 4.6 and 4.7 we show that these conditions control ramification in the extensions $K(\beta) \subseteq K_n(\beta)$, with consequences for the Galois theory of these extensions. We begin with the following standard lemma from Galois theory (see also [BT19, Lemma 6.1]).

Lemma 4.1. *Let $F_1, \ldots, F_n$ and M be fields all contained in some larger field. Assume that $F_1, \ldots, F_n$ are finite extensions of M.*

(i) If F_1 is Galois over M and $F_1 \cap F_2 = M$, then F_1F_2 is Galois over F_2 and $\mathrm{Gal}(F_1F_2/F_2) \cong \mathrm{Gal}(F_1/M)$.

(ii) If $F_1, \ldots, F_n$ are Galois over M with $F_i \cap \prod_{j \neq i} F_j = M$ for each i, then $\mathrm{Gal}(\Pi_{i=1}^n F_i/M) \cong \prod_{i=1}^n \mathrm{Gal}(F_i/M)$.

We prove another slightly more technical lemma that we will use throughout the rest of this paper.

Lemma 4.2. *Let M be a finite extension of a number field A.*

(i) Let F_1 and F_2 be finite extensions of M. Suppose that F_1 is Galois over M and that there is a prime $\mathfrak{p}$ of A such that $\mathfrak{p}$ does not ramify in F_2 but does ramify in any nontrivial extension of M contained in F_1. Then we have $\mathrm{Gal}(F_1F_2/F_2) \cong \mathrm{Gal}(F_1/M)$ and furthermore $\mathfrak{p}$ ramifies in any nontrivial extension of F_2 contained in F_1F_2.

(ii) Let $F_1, \dots, F_n$ be number fields that are all Galois over M. Suppose that for each F_i, there is a prime $\mathfrak{p}_i$ of A such that $\mathfrak{p}_i$ does not ramify in F_j for $i \neq j$ but does ramify in any nontrivial extension of M contained in F_i. Then $\mathrm{Gal}(\Pi_{i=1}^n F_i/M) \cong \prod_{i=1}^n \mathrm{Gal}(F_i/M)$ and furthermore any nontrivial extension of M that contained in $\Pi_{i=1}^n F_i$ must ramify over some $\mathfrak{p}_i$.

Proof. For (i), we note that we must have $F_1 \cap F_2 = M$ since $F_1 \cap F_2$ is unramified over $\mathfrak{p}$ and contained in F_1. Then by Lemma 4.1, we have $\mathrm{Gal}(F_1F_2/F_2) \cong \mathrm{Gal}(F_1/M)$. Thus, every extension E of F_2 contained in F_1F_2 has the form $E = M'F_2$ for some extension M' of M contained in F_1. If E is unramified over $\mathfrak{p}$, then we must have $M' = M$, by assumption, so E must equal F_2 as desired.

To prove (ii), note first that $F_i \cap \prod_{j\neq i} F_j = M$ for each i (since $\prod_{j\neq i} F_j$ is unramified over $\mathfrak{p}_i$), so $\mathrm{Gal}(\Pi_{i=1}^n F_i/M) \cong \prod_{i=1}^n \mathrm{Gal}(F_i/M)$, by Lemma 4.1. By part (i), every nontrivial extension of $\Pi_{j\neq i}F_j$ contained in $\Pi_{k=1}^n F_k$ ramifies over $\mathfrak{p}_i$. Thus, for each i, the group $G_i = \mathrm{Gal}(\Pi_{k=1}^n F_k/\Pi_{j\neq i}F_j)$ is generated by inertia groups of the form $I(\mathfrak{m}_i/\mathfrak{q}_i)$ where $\mathfrak{q}_i$ is a prime in $\Pi_{j\neq i}F_j$ lying over $\mathfrak{p}_i$. Note that each such $I(\mathfrak{m}_i/\mathfrak{q}_i)$ is contained in $I(\mathfrak{m}_i/\mathfrak{q}_i \cap M)$, because $\Pi_{j\neq i}F_j$ is unramified over $\mathfrak{p}_i$. Since the G_i generate $\mathrm{Gal}(\Pi_{i=1}^n F_i/M)$, this means that $\mathrm{Gal}(\Pi_{i=1}^n F_i/M)$ is generated by inertia groups over primes in M lying over the $\mathfrak{p}_i$. Thus, there can be no nontrivial extension of M contained in $\Pi_{i=1}^n F_i$ that is unramified over all $\mathfrak{p}_i$. □

We now define Conditions R and U. These are very similar to the definitions of Conditions R and U from [BT19].

Definition 4.3. Let $\beta \in \overline{K}$. We say that a prime $\mathfrak{p}$ of $K(\beta)$ satisfies **Condition R** at β for f and n if the following hold:

(a) f has good separable reduction at $\mathfrak{p}$;
(b) $v_\mathfrak{p}(f^i(0) - \beta) = 0$ for all $0 \leq i < n$;
(c) $v_\mathfrak{p}(f^n(0) - \beta)$ is positive and prime to p;
(d) $v_\mathfrak{p}(\beta) = 0$.

Definition 4.4. Let $\beta \in \overline{K}$. We say that a prime $\mathfrak{p}$ of $K(\beta)$ satisfies **Condition U** at β for f and n if the following hold:

(a) f has good separable reduction at $\mathfrak{p}$;
(b) $v_\mathfrak{p}(f^i(0) - \beta) = 0$ for all $0 \leq i \leq n$;
(c) $v_\mathfrak{p}(\beta) = 0$.

Remark 4.5. Note that if a prime $\mathfrak{p}$ of $K(\beta)$ satisfies Condition R at β for f and n, then it satisfies Condition U at β for f and $n-1$.

Proposition 4.6. *Let $\beta \in \overline{K}$. Let $\mathfrak{p}$ be a prime of $K(\beta)$ that satisfies Condition U at β for f and n. Then $\mathfrak{p}$ is unramified in $K_n(\beta)$.*

Proof. This is the content of [BT18, Proposition 3.1]. The proof in [BT18] is stated for $\beta \in K$, but works exactly the same if we allow $\beta \in \overline{K}$ and replace K with $K(\beta)$. □

The following result is similar to [BT19, Proposition 6.5].

Proposition 4.7. *Let $\beta \in \overline{K}$. Suppose that $\mathfrak{p}$ is a prime of $K(\beta)$ that satisfies Condition R at β for $n > 1$ and that $f^n(x) - \beta$ is irreducible over $K(\beta)$. Then*

$$\mathrm{Gal}(K_n(\beta)/K_{n-1}(\beta)) \cong C_q^{q^{n-1}}.$$

Furthermore, $\mathfrak{p}$ does not ramify in $K_{n-1}(\beta)$ and does ramify in any field E such that $K_{n-1}(\beta) \subsetneq E \subseteq K_n(\beta)$. Thus, we have

$$\mathrm{Gal}(M \cdot K_n(\beta)/M \cdot K_{n-1}(\beta)) \cong C_q^{q^{n-1}} \tag{4.7.1}$$

for any field M containing $K(\beta)$ that does not ramify over $\mathfrak{p}$.

Proof. Note that since K_1 contains a primitive q-th root of unity, and $n > 1$, the field K_{n-1} contains a primitive q-th root of unity. Recall that Condition R at β for n implies Condition U at β for $n-1$. By Proposition 4.6, $\mathfrak{p}$ does not ramify in $K_{n-1}(\beta)$.

Let $\bar{z}$ denote the image of $z \in \mathbb{P}^1(\overline{K})$ under the reduction mod $\mathfrak{p}$ map, which is well defined as long as $v_\mathfrak{p}(z) \geq 0$. Consider the map $\bar{f} : \mathbb{P}^1(\overline{k}_\mathfrak{p}) \to \mathbb{P}^1(\overline{k}_\mathfrak{p})$ that comes from reducing f at $\mathfrak{p}$, and recall that Condition R assumes that f has good reduction at $\mathfrak{p}$. We let $\bar{f}$ denote the reduction of f modulo $\mathfrak{p}$ and let $\bar{\beta}$ denote the reduction of β modulo $\mathfrak{p}$ as before. Since 0 is the only critical point of $\bar{f}$, it follows from (b) of Condition R that there are no critical points of $\bar{f}^{n-1}$ in $\bar{f}^{-(n-1)}(\bar{\beta})$. By (c) of Condition R, we see that $0 \in \bar{f}^{-n}(\bar{\beta})$, and that 0 is totally ramified over $\bar{f}(0) = \bar{c}$ (in the sense of $\bar{f}$ as a morphism from $\mathbb{P}^1(\overline{k}_\mathfrak{p})$ to itself). So $\bar{f}^n(x) - \bar{\beta}$ has 0 as a root of multiplicity q, and has no other repeated roots.

Now let $z_1, \ldots, z_{q^{n-1}}$ be roots of $f^{n-1}(x) - \beta$ and M_i be the splitting field of $f(x) - z_i$ over $K_{n-1}(\beta)$. By definition,

$$f^n(x) - \beta = \prod_{i=1}^{q^{n-1}} (f(x) - z_i)$$

over $K_{n-1}(\beta)$. Let $\mathfrak{q}$ be a prime of $K_{n-1}(\beta)$ lying over $\mathfrak{p}$. Then $\mathfrak{q}$ does not ramify over $\mathfrak{p}$, so

$$v_\mathfrak{q}(f^n(0) - \beta) = v_\mathfrak{p}(f^n(0) - \beta) > 0.$$

Also, z_i's are different modulo $\mathfrak{q}$, since if $z_i \equiv z_j \pmod{\mathfrak{q}}$ for some $i \neq j$, then $\bar{z}_i = \bar{z}_j$ which contradicts that $\bar{f}^{n-1}(x) - \bar{\beta}$ has no repeated roots. Therefore, we may assume $v_\mathfrak{q}(f(0) - z_1) = v_\mathfrak{q}(f^n(0) - \beta)$ and $v_\mathfrak{q}(f(0) - z_i) = 0$ for all $i \neq 1$.

Since $v_\mathfrak{q}(f(0) - z_1) = v_\mathfrak{q}(f^n(0) - \beta) = v_\mathfrak{p}(f^n(0) - \beta)$ is prime to p, we have $\mathrm{Gal}(M_1/K_{n-1}(\beta)) \cong C_q$. On the other hand, $\mathfrak{q}$ does not ramify in M_j for any $j \neq 1$, because $v_\mathfrak{q}(z_j - f(0)) = 0$. It follows that

$$M_1 \cap \left(\prod_{j \neq 1} M_j \right) = K_{n-1}(\beta).$$

As $f^n(x)-\beta$ is irreducible over $K(\beta)$, it follows from the Capelli's lemma that $f^{n-1}(x)-\beta$ is irreducible over $K(\beta)$ as well. Therefore all of the z_i are Galois-conjugate. That is, for any $z_j \neq z_1$, there exists $\sigma \in G_{n-1}(\beta)$ such that $\sigma(z_1)=z_j$. Applying σ to $\mathfrak{q}$, we obtain a prime $\sigma(\mathfrak{q})$ of $K_{n-1}(\beta)$ that ramifies in M_j with ramification index q and does not ramify in M_k for any $k \neq j$. Repeating the same argument as above, it follows that $\mathrm{Gal}(M_j/K_{n-1}(\beta)) \cong C_q$. Since for each j, there is a prime of $K_{n-1}(\beta))$ that ramifies completely in M_j but not in M_k for any $j \neq k$, so by part (ii) of Lemma 4.2 (with A and M taken to be $K_{n-1}(\beta)$ and the M_i taken to be the F_i), we have $\mathrm{Gal}(K_n(\beta)/K_{n-1}(\beta)) \cong C_q^{q^{n-1}}$. Likewise, by part (ii) of Lemma 4.2, every nontrivial extension of $K_{n-1}(\beta)$ contained in $K_n(\beta)$ must ramify over some $\sigma(\mathfrak{q})$ and thus over $\mathfrak{p}$.

□

Remark 4.8. One might hope that Proposition 4.7 holds under a natural weakening of Condition R where instead of requiring that $v_\mathfrak{p}(f^i(0)-\beta)=0$ for all $0 \leq i < n$ one only requires that $v_\mathfrak{p}(f^i(0)-\beta)$ be nonnegative and prime to p for all $0 \leq i < n$. However, this is not the case. Consider the case of $p=2$ with $f(x)=x^2-6$ and $\beta=111$ with the prime $\mathfrak{p}=(3)$ in $\mathbb{Z}$. Then $v_\mathfrak{p}(f(0)-\beta)=2$, $v_\mathfrak{p}(f^2(0)-\beta)=4$, and $v_\mathfrak{p}(f^3(0-\beta)=3$. One can calculate that $\mathfrak{p}$ does not ramify in $K_1(\beta)$, ramifies with index 2 in $K_2(\beta)$, and ramifies with index 4 in $K_3(\beta)$. Moreover for each prime $\mathfrak{q}$ in $K_2(\beta)$ lying over $\mathfrak{p}$, one can see that $K_3(\beta)$ contains a nontrivial extension of $K_2(\beta)$ that is unramified over $\mathfrak{q}$. Thus, the condition that $v_\mathfrak{p}(f^i(0)-\beta)=0$ for all $0 \leq i < n$ is necessary in Proposition 4.7.

With this notation we have the following result, which is similar to [BT19, Proposition 6.7]. First we need a little notation extending our earlier notation. Let $\boldsymbol{\alpha}=(\alpha_1,\dots,\alpha_s) \in \overline{K}^s$ and $L=K(\alpha_1,\dots,\alpha_s)$. We let $K_n(\boldsymbol{\alpha})$ denote the compositum $K_n(\alpha_1)\cdots K_n(\alpha_s)$. We let $G_n(\boldsymbol{\alpha})$ denote $\mathrm{Gal}(K_n(\boldsymbol{\alpha})/L)$ and let $G_\infty(\boldsymbol{\alpha})$ be the inverse limit of the $G_n(\boldsymbol{\alpha})$.

Proposition 4.9. *Let $\boldsymbol{\alpha}=(\alpha_1,\dots,\alpha_s)$ and L be the same as above and $n>0$. Suppose there exist primes $\mathfrak{q}_1,\dots,\mathfrak{q}_s$ of L such that*

(a) $\mathfrak{q}_i \cap K(\alpha_i)$ satisfies Condition R at α_i for f and n;
(b) $\mathfrak{q}_i \cap K(\alpha_j)$ satisfies Condition U at α_j for f and n for all $j \neq i$;
(c) $\mathfrak{q}_i \cap K$ does not ramify in L; and
(d) $f^n(x)-\alpha_i$ is irreducible over $K(\alpha_i)$ for all $i=1,\dots,s$.

Then $\mathrm{Gal}(K_n(\boldsymbol{\alpha})/K_{n-1}(\boldsymbol{\alpha})) \cong C_q^{sq^{n-1}}$. Furthermore, for any field E with $K_{n-1}(\boldsymbol{\alpha}) \subsetneq E \subset K_n(\boldsymbol{\alpha})$, there is an i such that $\mathfrak{q}_i \cap K$ ramifies in E.

Proof. For each $1 \leq i \leq s$, by (a), (d), and Proposition 4.7 (with $\mathfrak{q}_i \cap K$ playing the role of $\mathfrak{p}$ in the statement) we have

$$\mathrm{Gal}(K_n(\alpha_i)/K_{n-1}(\alpha_i)) \cong C_q^{q^{n-1}}.$$

We also have that every nontrivial extension of $K_{n-1}(\alpha_i)$ contained in $K_n(\alpha_i)$ must ramify over $\mathfrak{q}_i \cap K$.

Now for each $1 \leq i \leq s$, let $\mathfrak{p}_i = \mathfrak{q}_i \cap K$ and $L_i = K_n(\alpha_i) \cdot K_{n-1}(\boldsymbol{\alpha})$. By (a), (b), (c), and Proposition 4.6, the prime $\mathfrak{p}_i$ does not ramify in L_j for all $j \neq i$ and also does not ramify in $K_{n-1}(\boldsymbol{\alpha})$. By part (i) of Lemma 4.2, we see that every nontrivial extension of $K_{n-1}(\boldsymbol{\alpha})$ of contained in L_i must ramify over $\mathfrak{p}_i$. Thus, we may apply part (ii) of Lemma 4.2 (with the field M as $K_{n-1}(\boldsymbol{\alpha})$, the field A as K, and the fields F_i as L_i) to obtain

$$\begin{aligned}\mathrm{Gal}\left(K_n(\boldsymbol{\alpha})/K_{n-1}(\boldsymbol{\alpha})\right) &\cong \prod_{i=1}^{s} \mathrm{Gal}\left(L_i/K_{n-1}(\boldsymbol{\alpha})\right) \\ &\cong \prod_{i=1}^{s} \mathrm{Gal}\left(K_n(\alpha_i)/K_{n-1}(\alpha_i)\right) \cong C_q^{sq^{n-1}}.\end{aligned}$$

It also follows from part (ii) of Lemma 4.2 that every nontrivial extension of $K_{n-1}(\boldsymbol{\alpha})$ in $K_n(\boldsymbol{\alpha})$ must ramify over some $\mathfrak{p}_i$.

□

5. Proof of Main Theorems

The proofs of the main theorems combine the preliminary arguments from throughout the paper with the following proposition, which uses height arguments to produce primes with certain ramification behavior in $K_n(\beta)$. Recall the definitions of Condition R and Condition U from Section 4.

Proposition 5.1. *Let K be a number field, $q = p^r$ be a power of a prime number p, and $f(x) = x^q + c \in K[x]$ be a polynomial that is not PCF. Let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_s)$ be distinct strictly preperiodic points of f such that for each $i \neq j$ there is no $\ell \geq 0$ such that $f^\ell(\alpha_i) = \alpha_j$. Then there is an n_0 such that for all $n \geq n_0$, there are primes $\mathfrak{q}_1, \ldots \mathfrak{q}_s$ of $L = K(\alpha_1, \ldots, \alpha_s)$ such that*

(i) for each i, we have that $\mathfrak{q}_i \cap K(\alpha_i)$ satisfies Condition R at α_i for n;
(ii) for each $i \neq j$, we have that $\mathfrak{q}_i \cap K(\alpha_j)$ satisfies Condition U at α_j and f for all $m \geq 0$;
(iii) $\mathfrak{q}_i \cap K$ does not ramify in L.

Proof. Let $\mathcal{S}$ be a set of primes in L that includes all primes of L of bad or inseparable reduction for f, all primes $\mathfrak{q}$ of L such that $v_{\mathfrak{p}}(\alpha_i) \neq 0$ for some i (note that none of the α_i can equal to 0 since 0 is not preperiodic under f), all primes $\mathfrak{q}$ of L such that $\mathfrak{q} \cap K$ ramifies in L, and all primes of L such that $f^\ell(\alpha_i) \equiv \alpha_j \pmod{\mathfrak{q}}$ for some $\ell \geq 0$ and some $i \neq j$. By our assumptions, $\mathcal{S}$ is a finite set. Now for each i, let

$$\mathcal{S}_i = \{\mathfrak{q} \cap K(\alpha_i) : \mathfrak{q} \in \mathcal{S}\}.$$

By Proposition 3.1, for all sufficiently large n, there is a prime $\mathfrak{p}_i \notin \mathcal{S}_i$ of $K(\alpha_i)$ that satisfies Condition R at α_i for n. For each i, choose a prime $\mathfrak{q}_i$ of L lying over $\mathfrak{p}_i$. Then $\mathfrak{q}_i$ satisfies condition (i) and (iii).

Since $\mathfrak{q}_i \notin \mathcal{S}$, there is no $\ell \geq 0$ such that $f^\ell(\alpha_j) \equiv \alpha_i \pmod{\mathfrak{q}_i}$; it follows that we cannot have $f^m(0) \equiv \alpha_j \pmod{\mathfrak{q}_i \cap K(\alpha_j)}$ for any $m \geq 0$, since otherwise we would have

$$f^{n-m}(\alpha_j) \equiv f^n(0) \equiv \alpha_i \pmod{\mathfrak{q}_i}$$

when $n \geq m$ or

$$f^{m-n}(\alpha_i) \equiv f^m(0) \equiv \alpha_j \pmod{\mathfrak{q}_i}$$

when $m \geq n$. Thus $\mathfrak{q}_i$ also satisfies condition (ii). □

We will use the following theorem of Jones and Levy [JL17]. It is a special case of their Theorem 1.3.

Theorem 5.2. *Let K be a number field and let $f(x) = x^q + c \in K[x]$ where $q = p^r$ for a prime p and $v_\mathfrak{p}(c) \geq 0$ for some prime $\mathfrak{p}$ lying over p. Then for any $\beta \in K$ that is not periodic under f, the pair (f, β) is eventually stable over K.*

The following is a slight generalization of Theorem 1.2.

Theorem 5.3. *Let K be a number field. Let $q = p^r$ $(r \geq 1)$ be a power of the prime number p, let $f(x) = x^q + c \in K[x]$, where $v_\mathfrak{p}(c) \geq 0$ for some prime $\mathfrak{p}$ in K lying over p, and let $\beta \in \overline{K}$ be strictly preperiodic for f. Suppose that 0 is not preperiodic under f. Then we have $[\mathbf{G}_\infty : G_\infty(\beta)] < \infty$.*

Proof. By extending K, we may assume that $\beta \in K$ and K has a primitive q-th root of unity. Then by Theorem 5.2, the pair (f, β) is eventually stable over K. From Proposition 2.4, there is some $N \geq 0$ such that for all $\alpha \in f^{-N}(\beta)$ and for all $n \geq 1$, the polynomial $f^n(x) - \alpha$ is irreducible over $K(\alpha)$, which implies condition (d) in Proposition 4.9. On the other hand, applying Proposition 5.1 to $\boldsymbol{\alpha} = f^{-N}(\beta)$, there is an n_0 such that for all $n \geq n_0$, there are primes $\mathfrak{q}_1, \dots, \mathfrak{q}_s$ of $L = K_N(\beta)$ satisfying conditions (i), (ii), and (iii) in Proposition 5.1, which imply conditions (a), (b), and (c) in Proposition 4.9. (Here $s = q^N$.) Therefore, we have

$$\mathrm{Gal}(K_{N+n}(\beta)/K_{N+n-1}(\beta)) \cong C_q^{sq^{n-1}} = C_q^{q^{N+n-1}}$$

for all $n \geq n_0$. By Proposition 2.1, we are done. □

Theorem 5.4. *Let K be a number field. Let $f(x) = x^q + c \in K[x]$ where $q = p^r$ is a power of a prime number p and $v_\mathfrak{p}(c) \geq 0$ for some prime $\mathfrak{p}$ of K lying over p. Suppose that 0 is not preperiodic under f. Let $\beta_1, \dots, \beta_t \in K$ be preperiodic under f and suppose that there are no distinct j, k with the property that $f^\ell(\beta_j) = \beta_k$ for some $\ell > 0$. For each j, let M_j denote $K_\infty(\beta_j)$. Then for each $j = 1, \dots, t$, we have*

$$\left[M_j \cap \left(\prod_{k \neq j} M_k \right) : K \right] < \infty.$$

Proof. Applying Proposition 2.4 to each β_j and take the maximum, we have some $N \geq 0$ such that for all $\alpha \in B = \bigcup_{j=1}^{t} f^{-N}(\beta_j)$ and for all $n \geq 1$, the polynomial $f^n(x) - \alpha$ is irreducible over $K(\alpha)$. Let n_0 be as in Proposition 5.1 for $\boldsymbol{\alpha} = B$.

Now fix a j. We claim that

$$M_j \cap \prod_{k \neq j} M_k \subseteq K_{N+n_0}(\beta_j)$$

so that it has a finite index over K. It suffices to show that

$$K_{N+n}(\beta_j) \cap \prod_{k \neq j} M_k \subseteq K_{N+n_0}(\beta_j)$$

for all $n > n_0$, which is equivalent to

$$\left[K_{N+n}(\beta_j) \cdot \prod_{k \neq j} M_k : K_{N+n_0}(\beta_j) \cdot \prod_{k \neq j} M_k \right] = [K_{N+n}(\beta_j) : K_{N+n_0}(\beta_j)]$$

for all $n > n_0$. Using induction, it will suffice to show that

$$\begin{aligned} &\left[K_{N+n}(\beta_j) \cdot \prod_{k \neq j} M_k : K_{N+n-1}(\beta_j) \cdot \prod_{k \neq j} M_k \right] \\ &= [K_{N+n}(\beta_j) : K_{N+n-1}(\beta_j)] \end{aligned} \tag{5.4.1}$$

for all $n > n_0$.

Let $\alpha_i \in f^{-N}(\beta_j)$. For each $n > n_0$, there is a prime $\mathfrak{q}_i$ in $L = K(B)$ corresponding to α_i, satisfying (i), (ii), and (iii) of Proposition 5.1. $\mathfrak{q}_i \cap K$ does not ramify in $\prod_{k \neq j} M_k$ due to (ii) and (iii) of Proposition 5.1 and Proposition 4.6. Also, $\mathfrak{q}_i \cap K$ does not ramify in $K_{N+n_0}(\beta_j)$ due to (i) and (iii) of Proposition 5.1 and Proposition 4.6. Therefore, by Proposition 4.7, we have that 5.4.1 holds and our proof is complete.

□

6. The multitree

In this section we introduce a generalization of trees, which we call multitrees, in order to give a pleasant interpretation of Theorem 1.3 in terms of a finite index statement. For our purposes, we can simplify the presentation of multitrees in [BT19, Section 11] by avoiding the use of stunted trees.

Let $f \in K(x)$ with $\deg f \geq 2$ and set $\boldsymbol{\alpha} = \{\alpha_1, \ldots, \alpha_s\} \subseteq K$. Define

$$\mathcal{M}_n(\boldsymbol{\alpha}) = \bigcup_{i=0}^{n} \bigcup_{j=1}^{s} f^{-i}(\alpha_j)$$

and

$$G_n(\boldsymbol{\alpha}) = \mathrm{Gal}(K(\mathcal{M}_n(\boldsymbol{\alpha}))/K(\boldsymbol{\alpha})).$$

We refer to $\mathcal{M}_n(\boldsymbol{\alpha})$ as a *multitree*. It can be pictured as the union of s distinct trees of level n, rooted at the α_i.

As $n \to \infty$, define the direct limit

$$\mathcal{M}_\infty(\boldsymbol{\alpha}) = \varinjlim \mathcal{M}_n(\boldsymbol{\alpha})$$

and the inverse limit

$$G_\infty(\boldsymbol{\alpha}) = \varprojlim G_n(\boldsymbol{\alpha})$$

just as in the single tree case. For each n, $G_n(\boldsymbol{\alpha})$ acts faithfully on $\mathcal{M}_n(\boldsymbol{\alpha})$ in the usual way. So there are injections $G_n(\boldsymbol{\alpha}) \hookrightarrow \mathrm{Aut}(\mathcal{M}_n(\boldsymbol{\alpha}))$, and thus an injection $G_\infty(\boldsymbol{\alpha}) \hookrightarrow \mathrm{Aut}(\mathcal{M}_\infty(\boldsymbol{\alpha}))$, where an automorphism of the multitree must fix each root α_i.

Suppose that the individual trees rooted at α_i are disjoint, and that each α_i is neither periodic nor postcritical for f. Then the automorphism group of the infinite multitree has the simple description

$$\mathrm{Aut}(\mathcal{M}_\infty(\boldsymbol{\alpha})) \cong \mathrm{Aut}(T_\infty^d)^s,$$

that is, the direct product of s copies of $\mathrm{Aut}(T_\infty^d)$. This group has a subgroup $([C_q]^\infty)^s$, which is the direct product of s copies of the permutation group given by the infinite iterated wreath product action of C_q on T_∞^d. If there are s different polynomial maps $f(x) = x^d + c_i$ that satisfy the hypotheses of Theorem 1.2, then it is easy to see that $G_\infty(\boldsymbol{\alpha})$ embeds into $([C_q]^\infty)^s$. Thus we may rephrase Theorem 1.3 as a finite index statement. It is most easily stated when K contains a q-th root of unity.

Theorem 6.1. *Let K be a number field that contains a q-th root of unity and suppose that $\alpha_i \in K$ are strictly preperiodic for f. Suppose that there are no $i \neq j$ with the property that $f^\ell(\alpha_i) = \alpha_j$ for some $\ell \geq 0$. Then*

$$[([C_q]^\infty)^s : G_\infty(\boldsymbol{\alpha})] < \infty.$$

Proof. The group $G_\infty(\boldsymbol{\alpha})$ equals $\mathrm{Gal}(\prod_{i=1}^s K_\infty(f, \alpha_i)/K)$, which has finite index in the direct product $G_\infty(f, \alpha_1) \times \cdots \times G_\infty(f, \alpha_s)$ by Theorem 1.3 and basic Galois theory. This group in turn has finite index in $([C_q]^\infty)^s$ by applying Theorem 1.2 to each G_∞ separately. □

Remark 6.2. For periodic α, the point α appears repeatedly as its own inverse image, so the natural tree for $G_\infty(\alpha)$ act on is the product of the rooted binary trees corresponding to the strictly preperiodic elements of $f^{-1}(\alpha)$, so Theorem 6.1 provides a finite index theorem for periodic points of $x^q + c$ as well.

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